

Session 17 Quiz - Multimodal AI

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 53648]

_____ AI is artificial intelligence that processes and reasons about multiple types of data at once - text, images, audio, and video - rather than relying on a single modality.

- A. Unimodal
- B. Multimodal
- C. Reinforcement
- D. Symbolic

Correct Answer: B

Explanation: Multimodal AI combines several modalities at once - think of watching a movie where you see action, hear dialogue, and read subtitles all together. Unimodal AI uses only one type of data; multimodal AI fuses several.

Q2 [Type: MCQ | Difficulty: Easy | QID : 53649]

A _____ is a single type of data, such as text, image, sound, video, or code. Multimodal AI works with several of these at once.

- A. Modality
- B. Encoder
- C. Waveform
- D. Token

Correct Answer: A

Explanation: "Modality" is the technical term for a type of data. Combining different modalities is exactly what makes a system multimodal - not just having lots of input from the same kind of source.

Q3 [Type: MCQ | Difficulty: Medium | QID : 53650]

After encoding text, images, and audio into numbers, multimodal AI places all those _____ into a shared mathematical space (the Joint Embedding Space) so that similar concepts from different modalities end up close together.

- A. Waveforms
- B. Pixels
- C. Vector Embeddings
- D. Tokens

Correct Answer: C

Explanation: Vector embeddings are lists of numbers that capture meaning. Alignment in the Joint Embedding Space is what lets the system recognise that the word "raccoon" and a photo of a raccoon are about the same thing, even though they started in different modalities.

Q4 [Type: MCQ | Difficulty: Easy | QID : 53651]

A spam filter that ONLY reads email text - and ignores attached images or audio - is an example of multimodal AI.

- A. True
- B. False

Correct Answer: B (False)

Explanation: A system that uses only one type of data is unimodal, not multimodal. The classic example: "A spam filter only reads email text. It does not know if there is a funny picture attached." Multimodal AI requires multiple modalities, not just multiple inputs. A common mix-up is thinking multimodal means "lots of data." It actually means "different types of data."

Q5 [Type: MCQ | Difficulty: Medium | QID : 53652]

Deepfakes - videos or audio of people saying or doing things they never actually did - are a real challenge raised by the rise of multimodal AI.

- A. True
- B. False

Correct Answer: A (True)

Explanation: Deepfakes are explicitly named as one of the five multimodal AI challenges, alongside alignment, computational cost, cross-modal bias, and data scarcity. Because multimodal systems can generate convincing synthetic media, they can also be misused to fabricate fake events - which is why human oversight and verification matter most when outputs look flawless.

Q6 [Type: MCQ | Difficulty: Medium | Cognitive: Critical Thinking | QID : 53653]

Because multimodal AI can combine text, images, audio, and video, it truly understands what it senses the way a human brain understands a movie.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Multimodal processing is "all math, not magic" - the AI does not understand like a human; it follows mathematical steps to find and connect patterns. Combining modalities makes pattern matching richer, but it does not turn it into human-style understanding. This is the same warning that applies to NLP, CV, and Voice AI - multimodal AI is more impressive on the surface, but it is the same machinery underneath.

Q7 [Type: MCQ | Difficulty: Easy | QID : 53654]

Which of the following best captures the three-stage framework that all multimodal AI systems follow - and that you have seen across NLP, Computer Vision, and Voice AI?

- A. Encode → Decode → Translate
- B. Predict → Verify → Submit
- C. Sense → Think → Act
- D. Tokenize → Lemmatize → Generate

Correct Answer: C

Explanation: Sense → Think → Act is the foundational framework introduced earlier in the course. The NLP, CV, and Voice AI frameworks (Break/See/Hear → Process/Understand → Generate/Act/Speak) are all specialised versions of this same idea. Multimodal AI applies it across several modalities at once.

Q8 [Type: MCQ | Difficulty: Easy | QID : 53655]

Which of the following is an example of a multimodal AI TASK?

- A. Image captioning
- B. Stemming a word
- C. POS tagging a sentence
- D. Removing stop words from a text

Correct Answer: A

Explanation: Image captioning bridges two modalities - visual (input) and text (output) - making it a multimodal task. Stemming, POS tagging, and stop-word removal are all single-modality (text only) operations from NLP.

Q9 [Type: MCQ | Difficulty: Medium | QID : 53656]

Inside the THINK stage of multimodal AI, what are the three main sub-steps in the correct order?

- A. Fuse → Align → Encode
- B. Align → Fuse → Encode
- C. Sense → Predict → Act
- D. Encode → Align → Fuse

Correct Answer: D

Explanation: First Encode: turn each modality (image, text, audio) into vector embeddings. Then Align: place those embeddings in the Joint Embedding Space so similar concepts cluster together. Then Fuse: combine the aligned embeddings into one rich representation the AI can act on.

Q10 [Type: MCQ | Difficulty: Medium | QID : 53657]

Which of the following is a named challenge for multimodal AI?

- A. Pixel resolution
- B. Stemming errors
- C. Computational cost
- D. Phoneme detection

Correct Answer: C

Explanation: The five named challenges are alignment, deepfakes, computational cost, cross-modal bias, and data scarcity. Multimodal models are larger and slower than unimodal ones because they must process several types of data at once. The other options belong to single-modality domains (CV, NLP, Voice AI).

Q11 [Type: MCQ | Difficulty: Medium | Cognitive: Critical Thinking | QID : 53658]

Why is the "Align" step essential in multimodal AI's THINK stage?

- A.** It maps similar concepts from different modalities to nearby embeddings, so text and image versions of "raccoon" mean the same thing
- B.** It compresses the file size of the input data so the model can process multiple modalities without using too much memory
- C.** It speeds up the AI's response time by reducing the number of calculations needed to generate an output across modalities
- D.** It removes background noise from audio inputs and visual clutter from images before the model begins processing them

Correct Answer: A

Explanation: Without alignment, the text encoding of "raccoon" and the image encoding of a raccoon would land in unrelated parts of the embedding space - the system would not know they refer to the same concept. Alignment is what lets fusion produce a coherent answer instead of nonsense.

Q12 [Type: MCQ | Difficulty: Hard | Cognitive: Critical Thinking | QID : 53659]

Which of the following is best described as a UNIMODAL system, not multimodal?

- A.** A retail search tool that takes a photo plus a written query ("similar shoes, but in red") and returns matching products
- B.** An autonomous vehicle that combines camera, radar, and lidar input to detect obstacles
- C.** A smart assistant that looks at a photo of a dish, reads your recipe request, and speaks the instructions aloud
- D.** A grammar checker that only reads and edits plain text input

Correct Answer: D

Explanation: A grammar checker that only consumes text is using a single modality, so it is unimodal. The other three examples each combine at least two modalities - image + text, multiple sensor types, or image + text + audio - making them multimodal.

Q13 [Type: MCQ | Difficulty: Medium | QID : 53660]

Which task involves an AI answering a specific text question about the contents of an image - for example, "What is floating above the lighthouse?"

- A. Image Captioning
- B. Speech Recognition
- C. Visual Question Answering
- D. Text-to-Image Generation

Correct Answer: C

Explanation: VQA requires the AI to understand both the image and a specific text question, then provide a relevant text answer. Image captioning produces a general description without a specific question; speech recognition handles audio; text-to-image generates an image from a description.

Q14 [Type: MCQ | Difficulty: Hard | Cognitive: Critical Thinking | QID : 53661]

A hospital deploys a multimodal AI that combines patient imaging, lab results, and clinical notes to suggest diagnoses. After six months, a senior physician notices that junior doctors are increasingly leading with the AI's suggestion and working backwards to justify it, rather than forming their own assessment first. What is the most important problem this pattern reveals?

- A. Junior doctors are anchoring their reasoning to the AI's output instead of developing independent clinical judgment
- B. The AI system is performing too well - when suggestions are consistently accurate, it is natural for doctors to defer to them
- C. The hospital should restrict AI access to senior physicians only - junior doctors need more experience before using AI tools
- D. The documentation process is the real issue - if junior doctors phrase their notes differently, the reasoning quality would be unaffected

Correct Answer: A

Explanation: AI is supposed to inform clinical judgment, not replace the formation of it. When doctors anchor to AI output first and reverse-engineer justifications, the human-in-the-loop becomes a rubber stamp - and the clinical reasoning skills that distinguish good doctors from bad ones never develop.

Q15 [Type: MCQ | Difficulty: Hard | Cognitive: Critical Thinking | QID : 53662]

A multimodal AI helps dermatologists prioritise their patient queue by predicting which cases are most likely to be malignant. An ethics board finds the false negative rate is three times higher for South Asian patients than for white patients. The team responds: "Overall accuracy is still 91%, which exceeds the benchmark for clinical deployment." What is the most important flaw in this reasoning?

- A.** Aggregate accuracy hides distribution - a 91% rate with higher misses for one group means harm is concentrated unfairly
- B.** The ethics board does not have authority to block deployment on accuracy grounds - that decision belongs to the clinical team and hospital leadership alone
- C.** The false negative rate difference is within normal statistical variance and does not indicate a meaningful gap in how the model serves different patient groups
- D.** Aggregate accuracy hides who bears the cost - higher miss rates for one demographic mean errors are concentrated there

Correct Answer: D

Explanation: Overall accuracy is an average, and averages hide distribution. When errors cluster along demographic lines, the system is not performing at 91% for everyone - it is performing much worse for some patients while being inflated by better performance for others. A missed melanoma is not a statistic; it is a patient.

Q16 [Type: MCQ | Difficulty: Medium | QID : 53663]

A self-driving car combines input from its cameras, its radar, and its lidar to detect lane lines, a pedestrian, and a stop sign - all at once. Which characteristic of this system makes it a clear example of multimodal AI?

- A.** It uses more than one camera mounted at different angles around the vehicle to capture the road
- B.** It fuses different sensor types (different modalities) into a single understanding of the scene
- C.** It is fast at processing visual input and making driving decisions before reaching obstacles on the road
- D.** It only uses text-based commands from the driver to navigate through traffic

Correct Answer: B

Explanation: Multimodal is not about having lots of cameras - it is about combining different types of data. Cameras (images), radar (radio reflections), and lidar (laser distance measurements) are different modalities, and the car fuses all three into one scene understanding.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 53664]

Scenario: A multimodal AI is given the prompt "Generate an image of a sloth wearing a party hat and holding a balloon." The image it produces shows the sloth and the party hat clearly, but the balloon is completely missing.

Question: Which step of the Think stage most likely broke down?

- A.** Sense - the system never received the prompt from the user and could not begin processing
- B.** Encode - the system could not read or recognise the word "balloon" at all during input processing
- C.** Fuse - the system encoded and aligned the concepts but could not combine them all into a single scene
- D.** Act - the system refused to draw anything and produced no image as output

Correct Answer: C

Explanation: The system clearly received and understood the request (a sloth and a hat appear), so Sense and most of Encode worked. The breakdown is in fusion - combining all the relevant concepts into one scene. Some concept combinations are harder to represent together than others.

Q18 [Type: MCQ Scenario | Difficulty: Hard | Cognitive: Critical Thinking | QID : 53666]

Scenario: A high-quality multimodal model is used to create a video of a famous politician saying words they never actually said. The video looks convincing enough to fool many viewers, and it spreads on social media.

Question: Which multimodal AI challenge is most clearly at work, and why does it matter beyond just being technically interesting?

- A.** Computational cost - the video took expensive servers to render, making it slow and resource-heavy to produce
- B.** Deepfakes - AI-generated fake media can spread misinformation and erode public trust in real evidence
- C.** Data scarcity - there was not enough training data of the politician's speech for the model to learn properly
- D.** Alignment - the audio was out of sync with the video, making the fake easier for viewers to spot

Correct Answer: B

Explanation: Deepfakes are different in kind from accuracy challenges. The system is not failing; it is succeeding at a task that can cause real harm. This is why human oversight matters most when outputs look flawless - convincing fakes can mislead viewers, damage reputations, and undermine trust in genuine media.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | Cognitive: Critical Thinking | QID : 53667]

Scenario: You type into a multimodal AI: "Generate an image of a flamingo wearing a detective hat and holding a magnifying glass." A new image appears showing exactly that.

Question: Walk through the Sense → Think → Act framework for this interaction. At the THINK stage, name and briefly explain all three sub-steps (Encode, Align, Fuse).

(Respond in 3–4 sentences.)

Model Answer: In the Sense stage, the AI takes in the text prompt as a sequence of tokens such as "flamingo," "detective hat," and "magnifying glass." In the Think stage it Encodes the prompt into vector embeddings (numerical representations), Aligns those embeddings in a Joint Embedding Space so the concepts from text land near matching visual concepts the model knows, and Fuses everything into a single unified representation of the whole scene. In the Act stage, the system uses that fused representation to generate a brand-new image of a flamingo in a detective hat holding a magnifying glass. Throughout, the AI is matching mathematical patterns across modalities, not actually picturing or understanding the bird the way a human would.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | Cognitive: Critical Thinking | QID :53668]

Scenario: A regional hospital network is considering deploying a multimodal AI system that combines patient imaging, lab results, vital signs, and clinical notes to flag patients at risk of deterioration during their hospital stay. The CEO says this could save lives by catching warning signs nurses are too busy to notice. The nursing staff are split - some welcome the support, others worry about being held responsible for AI-flagged cases they do not have time to investigate or that the AI got wrong.

Question: Identify one significant benefit AND one significant risk of deploying this multimodal AI, and describe what the hospital should require to ensure the system genuinely helps patients and clinical staff.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that multimodal AI can identify subtle patterns across imaging, labs, vitals, and notes that may signal deterioration earlier than human clinicians monitoring single data sources can catch - potentially saving lives, especially when nursing staff are stretched thin. A significant risk is that AI-flagged alerts can create alert fatigue, lead to false positives that overwhelm nursing capacity, and shift legal and emotional responsibility onto nurses for AI's mistakes - and the AI's accuracy may not be equal across all patient populations, particularly those underrepresented in training data. To deploy responsibly, the hospital should require validation of the system's performance across diverse patient populations before deployment, establish clear protocols for how nurses respond to alerts, provide additional nursing capacity rather than expecting current staff to absorb the alert workload, and explicitly clarify that nurses are not professionally responsible for outcomes determined by AI errors.